

TrES-1b

R-2360

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_120711 · created 2026-08-19T23:25:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456119.9147 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.053
Transit depth (Rp/Rs)^2	0.00017 +/- 0.014 [%]
Orbital Inclination (inc)	86.93 +/- 1.46
Airmass coefficient 1 (a1)	2.85 +/- 2.46
Airmass coefficient 2 (a2)	0.33 +/- 0.66
Scatter in the residuals of the lightcurve fit is	92.21 %
Best Comparison Star	#1 - [184, 99]
Optimal Aperture	1.78
Optimal Annulus	4.84
Transit Duration (day)	0.074 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.053 vs 0.1358 (deviation 1.94σ)
Duration fit vs published	0.074 d vs 0.1042 d (deviation 1.05σ)
Mid-time O–C	-0.5 min
Depth SNR	0.0
Residual scatter	92.21 %

Light curve

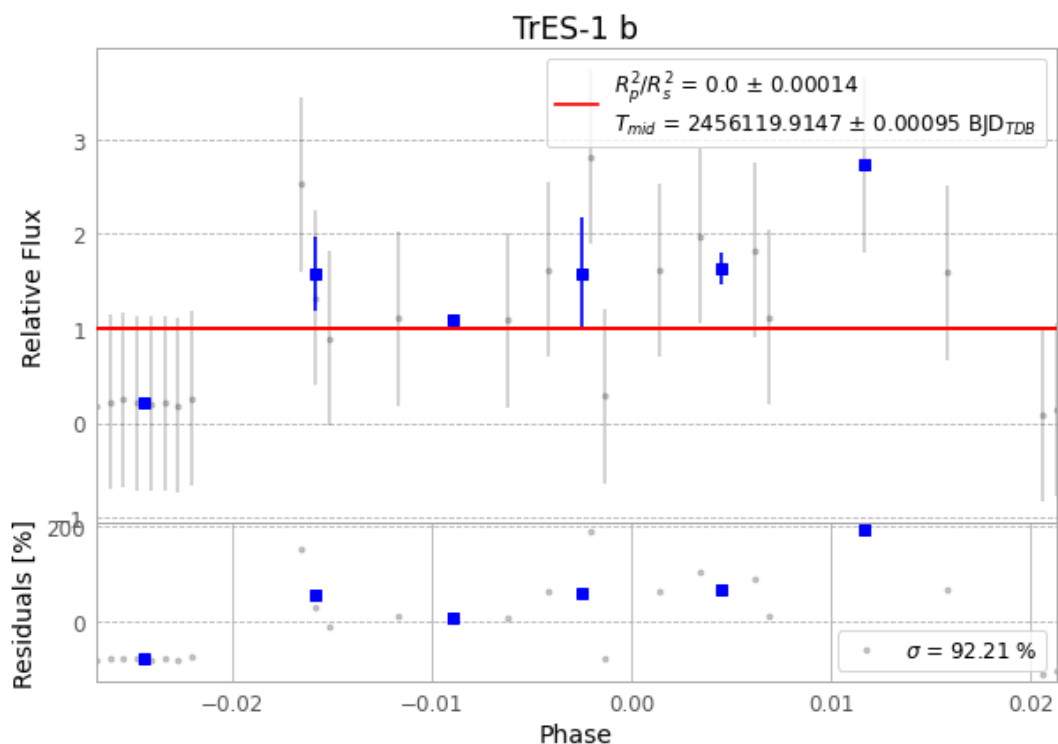


Figure 1 — FinalLightCurve_TrES-1 b_2012-07-11.png (SHA-256 0ebe1e96447c7f25...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 299], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 28569d376b1a8e49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), TRES-1120711075413.FITS → TRES-1120711112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-120711.log (88ce5ba979d4...), FinalLightCurve_TrES-1 b_2012-07-11.png (0ebe1e96447c...), FinalLightCurve_TrES-1 b_2012-07-11.csv (a74b0a3d60a6...)

Compute resources

Wall time 456.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:25Z.